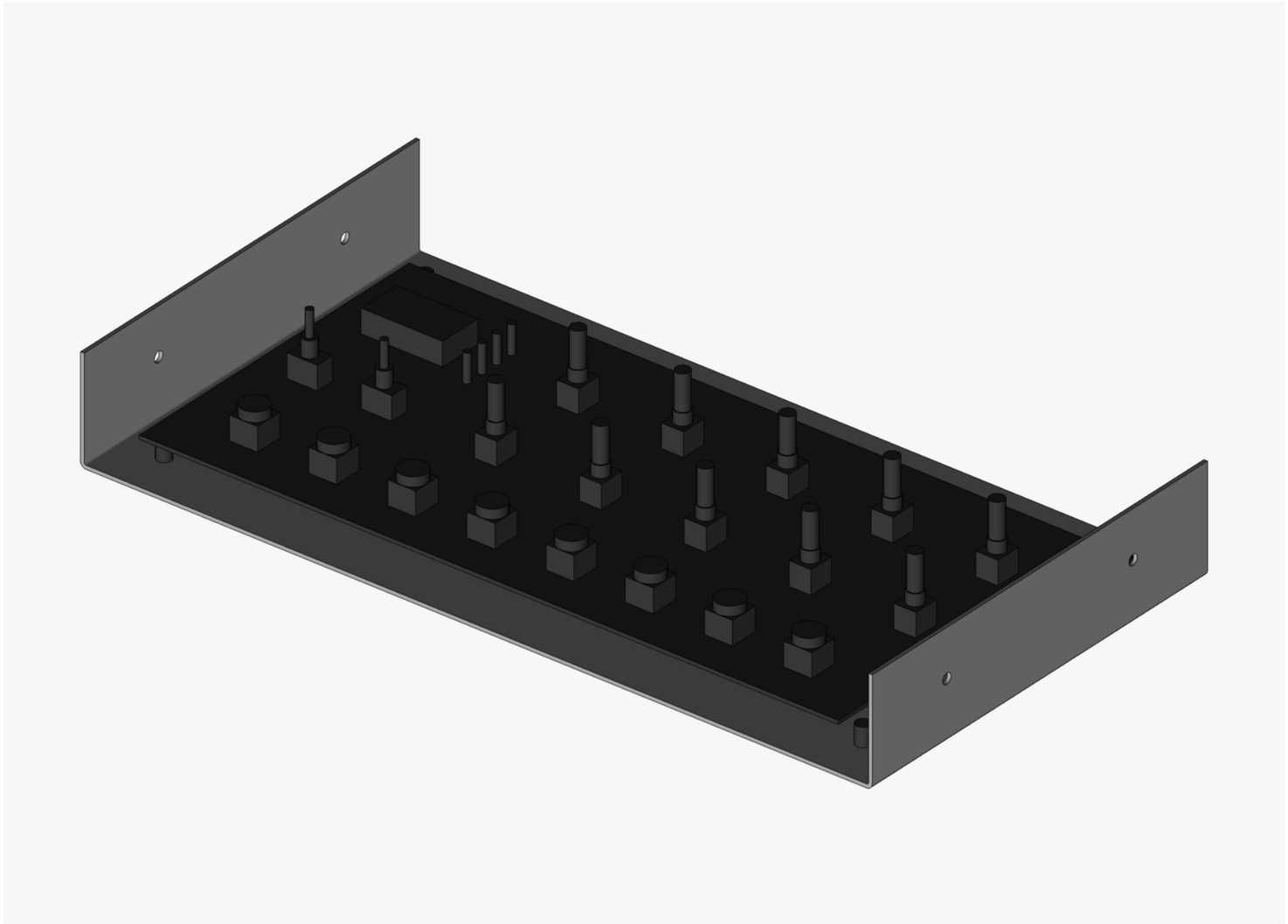


Checks: 20 pass, 0 warn, 0 fail. All 20 are on the last sheet.

Every check passes.

Sizes

Size	mm
Overall, everything	340 × 180 × 45
chassis, formed	340 × 180 × 45
chassis, flat	425.12 × 180

Parts list

#	Component	Qty	Source
1	chassis	1	sheet metal
2	standoff_M3x8	4	stand-in
3	control_board_pcb	1	STEP: control_board.step
4	oled_096	1	STEP: control_board.step
5	led_3mm	4	STEP: control_board.step
6	pot_9mm	10	STEP: control_board.step
7	toggle_m6	2	STEP: control_board.step
8	button_12mm	8	STEP: control_board.step

Parameters

Parameter	Value
shop	sendcutsend
material	5052 Aluminum
thickness	1.6
W	340
D	180
H	45
standoff	8

NOTES

1. Inside bend radius R0.889, K-factor 0.455.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. K 0.455 reproduces BD90 0.096 in (printed K 0.42).
5. Max bend length 44 in (1118 mm).

Chassis: assembly

qfold

PART	SCALE	SHEET
ASSEMBLY	NTS	1 of 4
MATERIAL	UNITS	REV / DATE
SendCutSend 5052 Aluminum, t=1.6	mm	A 2026-11-09

Formed size (outside) 340 × 180 × 45



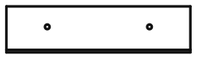
FLAT PATTERN 425.12 × 180



TOP (1:6)



FRONT (1:6)



RIGHT (1:6)

Holes (flat pattern, from lower-left)

Tag	Feature	X	Y	Size
M1	standoff screw M3	362.56	20	Ø3.4
M2	standoff screw M3	62.56	20	Ø3.4
M3	standoff screw M3	362.56	160	Ø3.4
M4	standoff screw M3	62.56	160	Ø3.4
E1	rack ear M4	20	40	Ø4.5
E2	rack ear M4	20	140	Ø4.5
E3	rack ear M4	405.12	40	Ø4.5
E4	rack ear M4	405.12	140	Ø4.5

Bends

Tag	Bend	Dir	Angle	In. R	Length	Allow.	Centre line
B1	right	DN	90°	R0.889	180	2.54	X 43.78
B2	left	DN	90°	R0.889	180	2.54	X 381.34

NOTES

1. Inside bend radius R0.889, K-factor 0.455.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. K 0.455 reproduces BD90 0.096 in (printed K 0.42).
5. Max bend length 44 in (1118 mm).
6. Flat pattern: outside face shown. DN = bend away from viewer. The DXF is the cutting master; the STEP is the formed part.

chassis		qfold
PART **CHASSIS**	SCALE 1:2	SHEET 2 of 4
MATERIAL SendCutSend 5052 Aluminum, t=1.6	UNITS mm	REV / DATE A 2026-11-09

Part	Tag	Feature	Size	X	Y	Z	Face	Facing
chassis	M1	standoff screw M3	Ø3.4	20	20	0	base	−Z down
chassis	M2	standoff screw M3	Ø3.4	320	20	0	base	−Z down
chassis	M3	standoff screw M3	Ø3.4	20	160	0	base	−Z down
chassis	M4	standoff screw M3	Ø3.4	320	160	0	base	−Z down
chassis	E1	rack ear M4	Ø4.5	340	40	25	right_wall	+X right
chassis	E2	rack ear M4	Ø4.5	340	140	25	right_wall	+X right
chassis	E3	rack ear M4	Ø4.5	0	40	25	left_wall	−X left
chassis	E4	rack ear M4	Ø4.5	0	140	25	left_wall	−X left

NOTES

1. Hole centres on the outside surface, in assembly coordinates.
2. Facing: the direction the hole opens to, out of the part.

Holes as assembled

qfold

PART	SCALE	SHEET
CHASSIS	NTS	3 of 4
MATERIAL	UNITS	REV / DATE
SendCutSend 5052 Aluminum, t=1.6	mm	A 2026-11-09

Checks: 20 pass, 0 warn, 0 fail

Check	Result	
chassis: valid solid	valid=True, solids=1, faces=22	PASS
chassis: no self-intersection	folded fuse loses 1.16e-10 mm³, flat fuse loses -2.91e-11 mm³ (limit 1.2e-04)	PASS
chassis: flat pattern matches fold	folded−flat volume 65.229828 mm³, bend allowance predicts 65.229828	PASS
chassis: hole to bend	tightest 15.81 mm (base standoff screw M3 to bend left), need ≥ 4.72	PASS
chassis: hole to edge	closest 15.81 mm (base standoff screw M3), need ≥ 1.60	PASS
chassis: web between holes	narrowest 95.50 mm (right_wall rack ear M4 / rack ear M4), need ≥ 1.60	PASS
chassis: holes inside their face	every hole lies within its face	PASS
chassis: minimum hole size	all holes ≥ 0.80 mm	PASS
chassis: bend corner relief	bends end ≥ 335.02 mm apart, need ≥ 1.60	PASS
chassis: flange length	shortest flat flange 42.51 mm (bend right), need ≥ 5.21	PASS
chassis.step reads back	1 solid(s), volume 122356.101 mm³ (model 122356.101)	PASS
assembly.step reads back	31 solid(s), volume 234002.409 mm³ (model 234002.409)	PASS
standoffs on the base	4 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
board on its standoffs	1 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
board clear of chassis	overlap 0.000000 mm³, closest 8.000 mm, need ≥ 3	PASS
faces of one part keep their gaps	1 unjoined face pairs, tightest 336.800 mm (chassis: right_wall vs left_wall), need ≥ 0.2	PASS
no part overlaps another	6 top-level parts, none overlapping	PASS
standoffs on their holes	4 parts, worst 0.000 mm across, 0.000 mm along, 0.000°	PASS
chassis: fits sheet	flat 425.1 × 180.0 mm, sheet limit 1117.6 × 762	PASS
chassis_flat.dxf reads back	8 circles, 4 lines, 0 arcs, 2 bends, cut 1309.5 mm, extent 425.123 × 180.000	PASS

NOTES

1. Run on every build; also in report.json.
2. Limits come from the project's rules (a shop's, or qfold's defaults in dfm.py).

Checks		qfold
PART **CHASSIS**	SCALE NTS	SHEET 4 of 4
MATERIAL SendCutSend 5052 Aluminum, t=1.6	UNITS mm	REV / DATE A 2026-11-09